



International Oil Supply, Trading, and Market Dynamics



Petroleum Economics and Decision Analysis

In the “perfect storm” of the energy commodity markets, oil trading is a business of decisions under uncertainty. This course builds the analytical foundation needed to understand crude oil markets, price behavior, and value drivers across the upstream–downstream value chain.

Participants explore how supply, demand, refining, transportation, and pricing mechanisms shape market outcomes. They examine how uncertainty, risk, and market structure influence trading positions and valuations. Through practical tools (including cash-flow logic, pricing models, derivatives, hedging strategies, and simulation) they learn how to quantify exposure, evaluate opportunities, and make disciplined trading and risk-management decisions in volatile energy markets.



LOCATION

Houston, USA

London, UK

Kuala Lumpur, Malaysia

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This course covers the analytical mindset for making informed trading and risk management decisions. It particularly aims to achieve these objectives:

- Identify key structures of crude oil and product markets.
- Explain price formation, refining margins, and market drivers.
- Apply trading instruments and valuation tools to assess exposure.
- Analyze uncertainty, risk, and market dynamics using simulation.
- Evaluate hedging strategies and trading decisions to maximize value.



This course is beneficial on organizational and personal levels. For organisations, this course

- Improves decision quality in trading and market analysis.
- Strengthens risk management and valuation capability.
- Enhances commercial performance through disciplined market insight.

From personal point of view, this course

- Builds a strong analytical and trading skillset.
- Increases confidence in managing uncertainty and exposure.
- Expands career opportunities in commercial and risk roles.

Course Objectives

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Day 1

Fundamentals of Crude Oil and Products Business

- Oil and Gas Industry—Value chain from subsurface to markets
- Energy Landscape and the future of oil and gas
- Global Crude Oil and Product Markets
- Reserve Base and Global Supply and Demand Forces
- Refinery Business and Products Markets
- Crude Oil Pricing Mechanisms
- Refinery Margins—Product Blending, Cracking, and Further Downstream and Petrochemicals
- Case Study: analysis of refinery planning, comparing refineries and their setups according to feedstocks

Day 2

Global Markets and Pricing Mechanisms

- Uncertainty and Risk in Energy Commodity Markets
- Supply, Transportation and Trading
- Market structure and regional enclaves: Role of OPEC and energy Geopolitics
- Light versus Heavy and Sweet versus Sour differentials
- Commodity Exchange, Their Benchmarks, and Market Instruments
- Spot Trading compared to Derivatives Trading
- Forwards, Futures, and Option Contracts
- Introduction to Monte Carlo Simulation
- Case Study: Using a simulation model for prices, we determine risk exposure and analyze trading positions.

Day 3

Risk Management in Energy Commodity Markets

- Risk in Financial Commodities Investments
- Hedging strategies for Risk Management
- Hedge-and-Forget versus Dynamic Hedging
- Convenience Yield in Futures Prices
- Contango and Backwardation in Oil and Gas Markets—opportunities for value creation
- Emerging Trends: Digital Technology, ESG, and Emission Markets
- Case study: Analyzing tools and techniques for risk management and maximizing value.

Contact us



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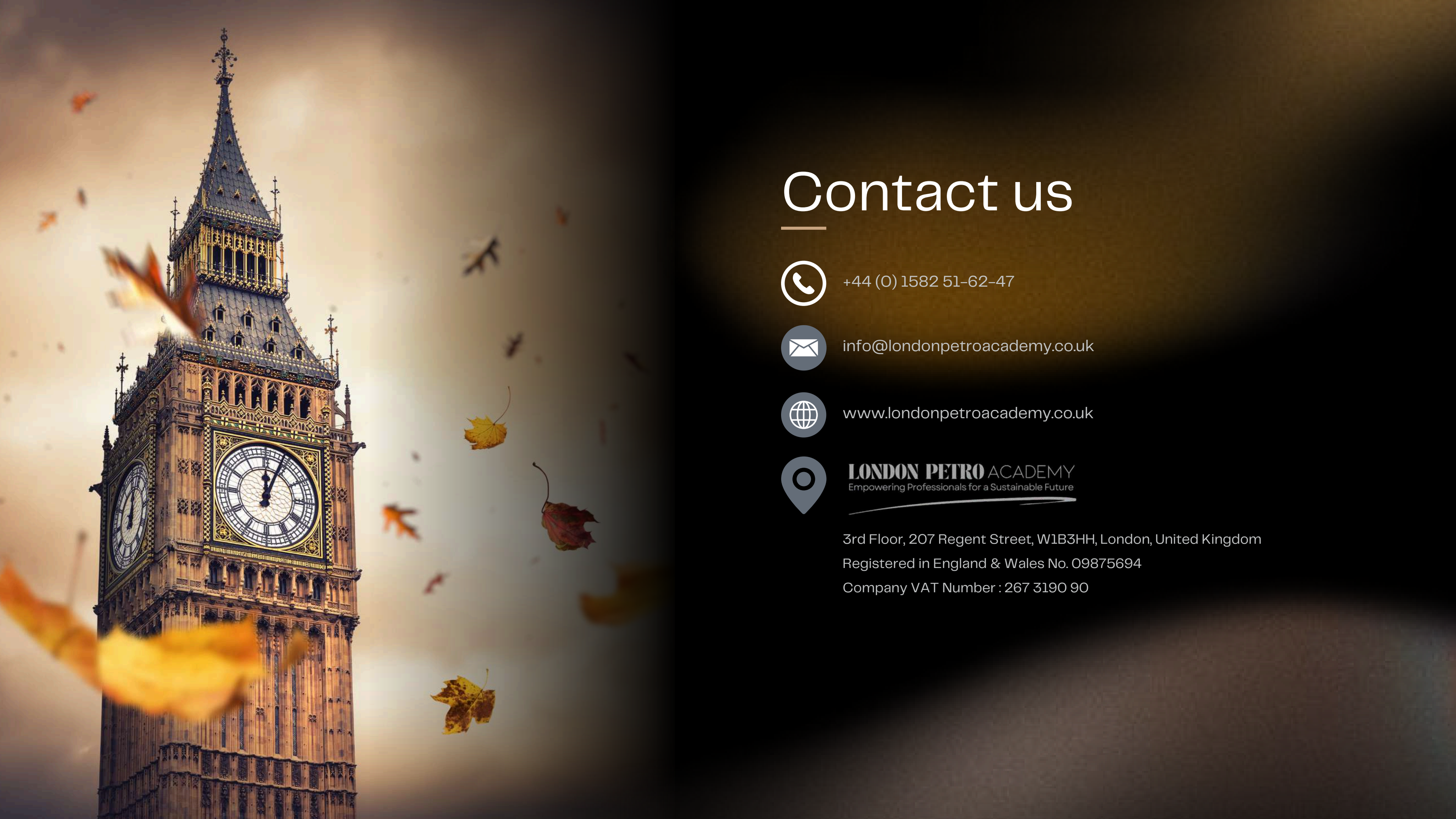
BABAK JAFARIZADEH

Dr. Babak Jafarizadeh is a senior consultant and engineering economist who views engineering as the pursuit of economic solutions to technical challenges in an uncertain world. With extensive experience in the oil and gas industry, he has served as a senior economic analyst supporting multi-million-dollar investment decisions for projects across the globe.

Since 2019, Dr. Jafarizadeh has worked as a senior consultant for leading training and consulting companies, while also holding academic appointments at several prestigious institutions — including Heriot-Watt University in the UK (Assistant Professor), the University of California, Berkeley (Lecturer at the Fung Institute for Engineering Leadership), and previously the University of Greenwich (Senior Lecturer). His earlier industry experience includes serving as a Senior Economic Analyst with Equinor (formerly Statoil) in Norway and the USA.

Dr. Jafarizadeh holds a PhD in Petroleum Engineering with a concentration in Investment and Decision Analysis from the University of Stavanger (in collaboration with the University of Texas at Austin), as well as an MSc in Financial Engineering and a BSc in Industrial Engineering. He is an active member of several professional associations, including the Society of Petroleum Engineers (SPE), the Institute for Operations Research and the Management Sciences (INFORMS), and the European Association of Geoscientists and Engineers (EAGE).

His teaching and research interests lie at the intersection of decision analysis and corporate finance, with particular applications in engineering and the energy industry. He has designed and delivered numerous professional training courses, such as Advanced Financial Modelling for Oil and Gas Professionals, Advanced Project Economics, Oil and Gas: Markets, Trading, Pricing and Economic Frameworks, and Energy Economics. Dr. Jafarizadeh is also the author of the book Economic Decision Analysis (for Project Feasibility Studies) and continues to contribute to both academia and industry through research, consulting, and executive education. His areas of expertise include petroleum economics, investment and decision analysis, project feasibility studies, corporate finance, and financial modelling for energy projects.



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